USER´S MANUAL

Thermia Duo
Heat Pump

9680-54962001 Rev. 2

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1 Important information

- If the installation will be shut down during the winter months, the heating system should be emptied of water to avoid any damage due to freezing. (Call your Installation Contractor for help, see "References" on page 32).
- The installation is practically maintenance free, but some monitoring will be required (see "Operating Instructions" on page 9).
- Before making any adjustments of system parameters, first find out, what the consequences are in each case.
- For all service work, please call your installation contractor.

1.1 Safety Instructions

1.1.1 Installation and maintenance

- Installation and commissioning as well as repairs and maintenance must be carried out by a qualified installation contractor (see "References" on p. 32).
- Any changes in the electrical installation must be undertaken by a qualified electrician (see "References"on page 32).



Danger to life!

- Service work on the refrigerant circuit must be carried out by a qualified refrigeration technician (see "References" on page 32).

1.1.2 Modification of System Components

The below components can only be modified by a qualified installation contractor:

- Heat pump unit
- Refrigerant, chilled water, warm water and power lines
- Safety valve

No structural changes may be made to the installation that might affect the operational reliability of the heat pump.

1.1.3 Safety valve

The following safety instructions must be observed in regard of the safety valve of the heating circuit and the overflow pipe:

- Never close the connection between safety valve and overflow pipe.
- When heated, water tends to expand which causes some water to be let out through the overflow pipe. This water may be hot! The overflow pipe should therefore discharge into a floor drain placed so that nobody can get hurt.

1.2 Protective Measures

1.2.1 Preventing Corrosion

To eliminate any risks of corrosion, the use of all kinds of aeorosol should be avoided in the proximity of the heat pump. In particular, avoid the following:

- Solvents
- Detergents containing chlorine
- Paint
- Adhesives

2 About your heat pump

2.1 Product description

Thermia Duo is a complete heating installation for providing heating and hot water if combined with an external warm water tank. It uses the first compressor on the market to be developed specifically for heat pumps. A new evaporator permits a new and improved refrigerant distribution system. It also has a built-in additional heater.
Thermia Duo is equipped with a control system that is controlled with the aid of a graphical character display. Thermia Duo Optimum is equipped with speed control. The heat pump's control computer reads off the actual temperatures of the heating, hot water and coolant system and adapts the speed of the circulation pumps accordingly. This means that your heat pump always adapts itself to the system that it is connected to. All to achieve the highest possible savings.
Heat is supplied to the house via a water-filled heating system, known as a low-temperature system, that delivers a maximum water temperature (supply line temperature) of 55°C to the radiators. The heat pump delivers as much of the heat demand as possible before switching in the additional heating to help meet the demand.
The Thermia Duo heating installation consists of four base units:

1 Heat pump unit

- Scroll compressor
- Stainless steel heat exchanger
- Circulation pumps for the collector system and heating system
- Valves and safety equipment for the cooling system and associated electrical components

2 Switching valve

- This opens and closes the connection to the warm water tank depending on whether heating or warm water are require

3 Additional heater

- 9 kW immersion heater
- Three-stage immersion heater control
- Installed in supply line
- Covers the need for additional heat if the capacity of the heat pump unit is exceeded
- Can be switched on automatically by the heat pump unit when the "AUTO" operating mode is selected

4 Control equipment

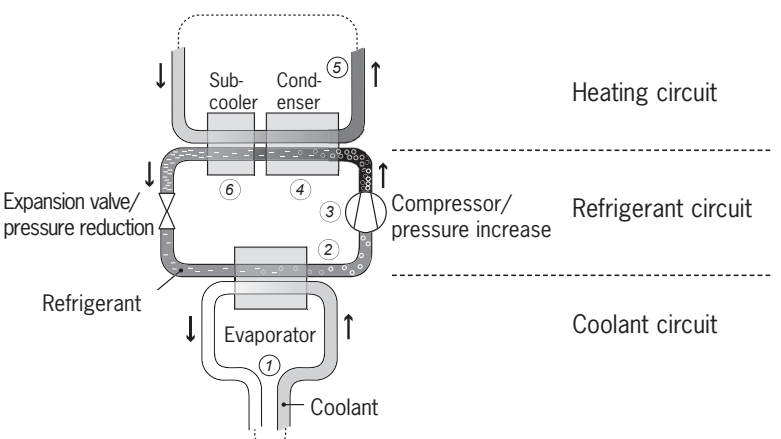
The control equipment controls the components of the heating installation (compressor, circulation pumps, additional heater and switching valve) and keeps track of when the heat pump is required to start and stop, and whether it is required to produce heat for the house or warm water.

- Control computer with graphical character display
- Temperature sensors (outdoor, supply line, return line, coolant and warm water)
- Room thermostat (option)

2.2 How a heat pump works

A heat pump makes use of the energy that exists in a natural heat source. In simple terms you could say that it extracts energy in the form of heat from a heat source.

The heat pump therefore has very low environmental impact and provides a very economical way of heating a house.



1 A hose filled with liquid (coolant) is submerged in a lake, buried under the ground or buried in bedrock. The liquid coolant absorbs energy from the heat source, since the temperature of the liquid in the hose is raised a few degrees by the heat source surrounding it.

2 The coolant is then fed into the heat pump's evaporator. Inside the evaporator the pressure is reduced, which forces the refrigerant to boil so that it changes into its gaseous form (vaporizes). The energy that is required for this process is given off by the coolant.

3 The refrigerant, which now contains a large amount of energy in the form of heat, is then pumped into the compressor, which raises its temperature and pressure.

4 The refrigerant then passes into the condenser. As it condenses, the refrigerant gives off its heat energy to the heating circuit. The temperature of the refrigerant drops, and it returns to its liquid form.

5 The heating circuit carries this heat energy to the warm water tank, radiator system and underfloor heating system, which are warmed up.

6 The refrigerant is still warm after passing through the condenser. In the sub-cooler the return water from the radiator system and underfloor heating system is pre-heated. The refrigerant is then led through the expansion valve, where the pressure is reduced and the whole process starts again.

The heat pump has three separate fluid circuits:

- **The coolant circuit** - a water-based mixture that carries energy from the heat source to the heat pump.
- **The refrigerant circuit** - the circuit in the heat pump that absorbs energy from the coolant circuit by means of vaporization, compression and condensation, and transfers it to the heating circuit. The refrigerant does not contain chlorine.
- **The heating circuit** - the water that transfers the heat energy to the heating system and the warm water tank.

2.3 Additional heating – immersion heater

Additional heating is provided by an immersion heater in the supply line that can be set for three different power output stages:

- Stage 1 = 3 kW
- Stage 2 = 6 kW
- Stage 3 = 9 kW

2.4 Warm water tank

The Thermia Duo can also supply warm water to an external warm water tank. The temperature of the water delivered to the warm water heater is controlled by the control pressostat and cannot be adjusted.

Warm water production is stopped when the control pressostat reaches its maximum working pressure, which is equivalent to a warm water temperature of around 50–55°C.

The water in the warm water tank is heated to 65°C at a preset time interval to prevent the growth of bacteria. The factory-set time interval is 14 days. For more information see "Warm water production" on page 23.

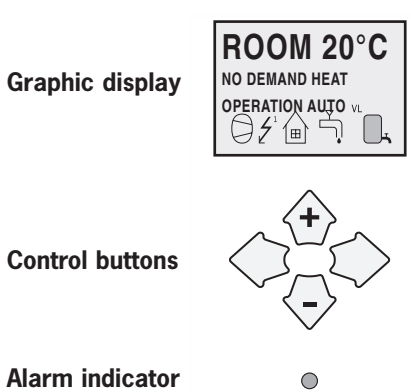
2.5 Room thermostat (option)

The heat pump system can be fitted with an optional room thermostat. This must be installed by an authorized installer. During normal operation the room temperature setting is displayed (e.g. 20.0). When a button is pressed, the set value is displayed in flashing characters and can then be adjusted using the + or – buttons. The outdoor temperature is displayed for 10 seconds if both buttons are pressed at the same time. The letter A is shown in the display if any of the alarms is triggered.

2.6 Control panel

The control panel of Thermia Duo features a graphic display, four control buttons and an alarm indicator.

Beside the control panel you will find the User's Manual, a short description of how to increase and reduce room temperature, and a label with name and phone number of Thermia distributors.



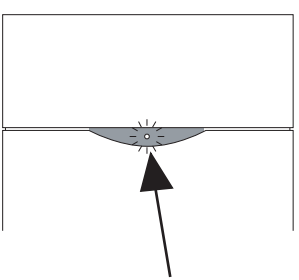
2.6.1 Functions

The control computer is operated with the help of a user friendly menu system that is displayed on the control panel. There is a main menu and four sub-menus accessible from the main menu. The menus are described in detail further down.

To be able to select the desired menu and increase or reduce preset values, you will use the four control buttons.

- One button pointing upwards marked with a plus
- One button pointing downwards marked with a minus.
- One button pointing to the right
- One button pointing to the left

The alarm indicator at the bottom of the control panel will tell you whether the unit is working properly (fixed light) or whether an alarm has been activated (blinking light). The indicator is clearly visible even if the upper panel is shut.



Alarm indicator

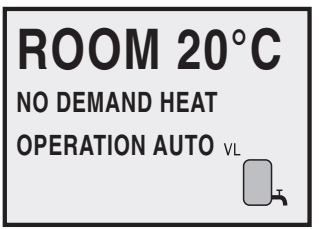
3 Operating instructions

3.1 General Information

3.1.1 Menu Navigation

The right-hand button on the control panel is used to open the desired menu. The left-hand button is used to return to the previous menu. The up and down buttons are used to navigate between the parameters of a menu. A cursor (arrow) on the left-hand side of the display indicates which menu can be opened. The up and down buttons are also used if you wish to increase or reduce a preset value.

3.1.1.1 Display of current operatingmode



During normal operation, the following information will be displayed:

- Desired (preset) room temperature
- Whether there is a heating demand or not. If there is, there will also be symbols telling which heat source is working – heat pump or auxiliary heater – or both (see "Symbols" on page 10).
- Which operating mode has been selected

In addition to this, the following messages may be displayed:

- **PHASE SEQUENCE**
  The compressor in the heat pump is running in the wrong direction, which means that heat is produced by the auxiliary heater only (see "Alarm" on p. 27).
- **HIGH RETURN**
  The temperature of the water returning from the radiators is too high so the heat pump will not be allowed to produce heat. (see "Alarm" on p. 27).
- **START**
  The heat pump is in the processs of starting. There is a 30 second delay between start of chilled water pump and compressor.
- **START_MIN**
  The heat pump will start in the number of minutes indicated in the display.

3.1.1.2 Symbols

To provide quick information about the operating status of the installation the following symbols are shown at the bottom of the character display depending on which part of the installation is operating:



This symbol shows that the heat pump is operating.



This symbol shows that additional heating is switched on. The figure next to the symbol shows the current power setting.



This symbol shows there is a demand for heat.



This symbol shows that warm water is being produced.



This symbol shows the status of warm water production. When the symbol is empty it shows that the warm water temperature is 44°C.



When the symbol is full it shows that the warm water temperature is 55°C.



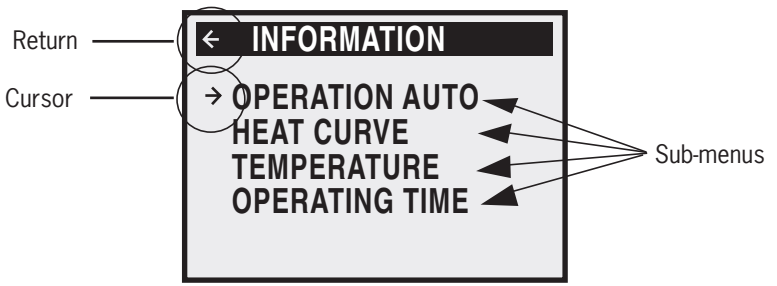
Shown if temperature reduction is switched on.

VL Shows which type of system the installation is set up for. It is set by the authorized installer during installation and cannot be changed.

3.1.2 Menus

3.1.2.1 Main menu "INFORMATION"

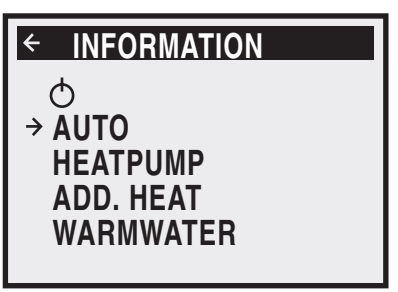
Pressing the right or left button once opens the control computer's main menu, "INFORMATION".



Use the up and down buttons to choose the desired sub-menu and open this menu by pressing the right button once. To return to the character display's starting point, press the left button once.

3.1.2.2 Sub-menu "OPERATION"

You use the OPERATION menu to select the operating mode for the heat pump.



For more information see "Setting the operating mode" on page 14.

3.1.2.3 Sub-menu "HEATCURVE"

This menu is used for making adjustments that affect the room temperature. For more information, please refer to "Adjustments to be made regularly" on page 14.

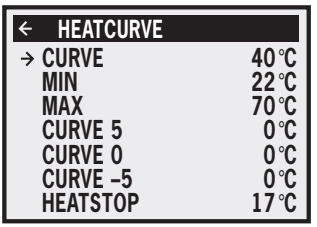


Table 1: Menu – HEAT CURVE

Table 1: Menu – HEAT CURVE

Menu text	Description	Adjusted by:
CURVE	The value entered shows the temperature of the water to be distributed to the radiators (feed line temperature) when the outside air temperature is 0°C.	User (see "Adjustment of CURVE value" on p. 16)
MIN	Adjustment of value for lowest feed line temperature allowed.	User (see "Adjustment of MIN and MAX values" on p. 20)
MAX	Adjustment of value for highest feed line temperature allowed.	User (see "Adjustment of MIN and MAX values" on p. 20)
CURVE 5	Adjustment of room temperature when the outside air temperature is +5°C.	User (see "Adjusting the "CURVE" setting" on page 16).
CURVE 0	Adjustment of room temperature when the outside air temperature is 0°C.	User (see "Adjusting the "CURVE" setting" on page 16).
CURVE - 5	Adjustment of room temperature when the outside air temperature is -5°C.	User (see "Adjusting the "CURVE" setting" on page 16).
HEAT STOP	Production of radiator heat is stopped when outside air temperature is equal to or higher than the value for heat stop entered.	User, if required.
REDUCTION	Room temperature can be reduced if an external timer is installed. The control system determines how much the temperature should be reduced.	User, if required.

3.1.2.4 Sub-menu “TEMPERATURE”

The "TEMPERATURE" menu shows the different temperatures that the installation has reached. The temperature is recorded continuously for the last 60 minutes, so it can also be displayed in the form of graphs.

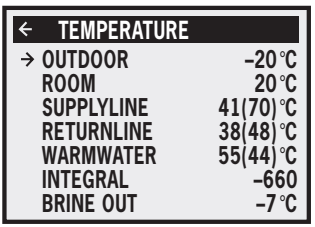


Table 2: Menu TEMPERATURE

Menu text	Description	Adjusted by:
OUTDOOR	Outdoor temperature	—
ROOM	Desired indoor temperature	User (see "Adjusting the "ROOM" setting” on page 17).
SUPPLYLINE	Shows the measured supply line temperature for the heating system. The set value is shown in brackets.	Not adjustable (the computer calculates what temperature is required to maintain the indoor temperature).
RETURLINE	Shows the measured return line temperature for the heating system. The maximum permissible temperature is shown in brackets.	Authorized installer. Adapted to suit individual installation.
WARMWATER	Shows measured warm water temperature. The start temperature for warm water production is shown in brackets.	Not adjustable.
INTEGRAL	Heat production is controlled by the estimated demand, which is calculated automatically while the installation is in operation. The integral is the energy balance of the heating system.	Not adjustable.
BRINE IN	The temperature of the coolant when it comes from the heat source.	Not adjustable.
BRINE UT	The temperature of the coolant when it returns to the heat source.	Not adjustable.

3.1.2.5 Sub-menu "OPERATING TIME"

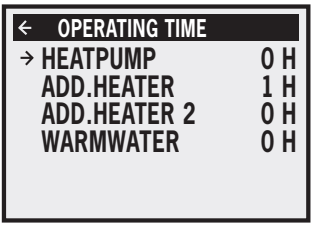


Table 3: Menu OPERATING TIME

Menu text	Description	Adjusted by:
HEATPUMP	Calculates the total time in hours that the heat pump has been in operation since it was installed, i.e. the operating time is not zeroed.	Not adjustable.
ADD.HEAT	Calculates the total time in hours that the additional heater (3 kW) has been in operation since it was installed, i.e. the operating time is not zeroed.	Not adjustable.
ADD.HEAT 2	Calculates the total time in hours that the additional heater (6 kW) has been in operation since it was installed, i.e. the operating time is not zeroed.	Not adjustable.
WARMWATER	Calculates the total time in hours that the warm water tank has been in operation since it was installed, i.e. the operating time is not zeroed.	Not adjustable.

3.2 Adjustments to be made regularly

When the heat pump is installed most settings are made by an authorized installer. The fine adjustments that you need to make afterwards are:

- Setting the operating mode
- Setting the desired room temperature using the "ROOM" setting
- Adjusting the heat curve
- Setting the maximum and minimum supply line temperatures

You can also change the heat stop setting, although this is not necessary. (See "Setting "HEAT STOP"" on page 21).

3.2.1 Setting the operating mode

The control computer lets you choose between the five following operating modes:

- **OPERATION OFF**
  The installation is switched off completely. The text "OPERATION ⏻" appears in the control panel's character display.



Remember that if the installation is switched to "OPERATION OFF" for long periods in the winter the heating system must be drained of water, otherwise there is a risk of pipes bursting due to freezing.

- **OPERATION AUTO**
  The heat pump and additional heater are controlled automatically by the control computer. The text "OPERATION AUTO" appears in the control panel's character display.
- **OPERATION HEATPUMP**
  The control computer only allows the heat pump unit (compressor) to operate. The text "OPERATION HEATPUMP" appears in the control panel's character display.
- **OPERATION ADD.HEAT**
  The control computer only allows the additional heater to operate. This operating mode can be used when a new installation is brought into use, if the coolant system is not ready for operation. The text "OPERATION ADD.HEAT" appears in the control panel's character display.
- **OPERATION WARMWATER**
  In this mode the heat pump only produces warm water; it does not provide any heat to the heating system. The text "OPERATION WARMWATER" appears in the control panel's character display.

To change the operating mode:

1 Press the right or left button once to select the main "INFORMATION" menu.
  The cursor will highlight the menu selection "OPERATION".

2 Open your choice by pressing the right button once.
  The cursor will highlight the operating mode that was used last.

3 Select the new operating mode using the up or down buttons.

4 Press the left button twice to return to the character display's operating mode.

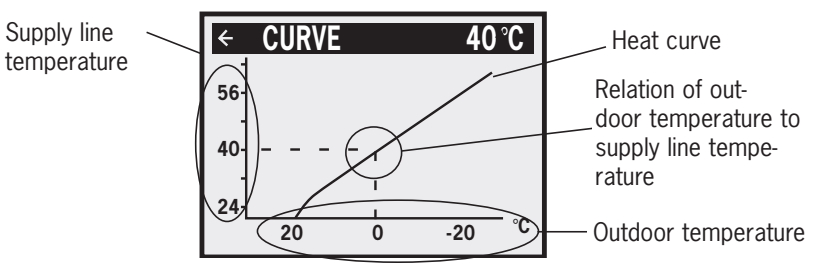
3.2.2 Heat production – general description

You adjust the indoor temperature by changing the heat curve for the installation. The heat curve is the tool used by the control computer to calculate the temperature of the supply line water that is delivered to the heating system.

The heat curve is set when the heat pump is installed, but it is important that it is adjusted afterwards so that the indoor temperature is comfortable in all weather conditions. When the heat curve is correctly adjusted it minimises the attention required and ensures energy-efficient operation.

The heat curve determines the temperature of the supply line water depending on the outdoor temperature. In other words, the colder it is outside, the higher the supply line temperature will be. This means that the supply line temperature delivered to the radiators increases exponentially as the outdoor temperature drops.

A graph is displayed when you choose "CURVE" from the "HEATCURVE" sub-menu. The graph shows the relationship between the supply line temperature and the outdoor temperature. It is the relationship between them that is called the heat



curve.

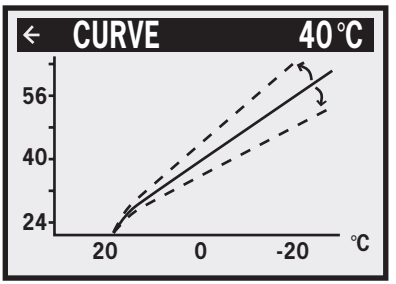
3.2.3 Adjusting the "CURVE" setting

You can adjust the heat curve by changing the "CURVE" setting. The "CURVE" setting determines the supply line temperature for the radiators when the outdoor temperature is 0°C. When the outdoor temperature is below 0°C the temperature of the supply line water that is piped to the radiators is raised above 40°C, and when the outdoor temperature is above 0°C the temperature of the supply line water is reduced below 40°C. When you increase the "CURVE" setting it makes the slope of the heat curve steeper, and when the setting is reduced it makes the slope shallower.

This method of adjusting the indoor temperature should be used to make a long-term change to the temperature, as this is the most energy-efficient and cost-effective method. A temporary increase or decrease can be made by adjusting the "ROOM" setting instead (see "Adjusting the "ROOM" setting" on page 17).

The factory setting is 40°C at an outdoor temperature of 0°C. The setting can be

Changing the CURVE setting



To change the "CURVE" setting:

1 Press the right or left button once to select the main "INFORMATION" menu.

  The cursor will highlight the menu selection "OPERATION"

2 Press the down button to move the cursor over the menu selection "HEATCURVE".

3 Open the menu by pressing once on the right button. The cursor will highlight "CURVE".

4 Open your selection by pressing the right button once.

5 Increase or decrease the setting using the up or down buttons.

  The graph will show how the slope of "CURVE" changes

Press the left button three times to return to the character display's operating mode.

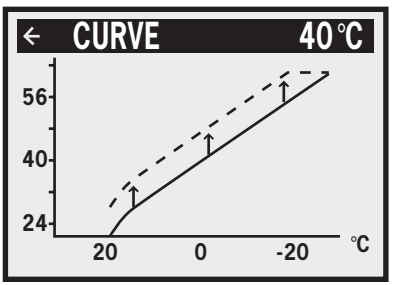
3.2.4 Adjusting the "ROOM" setting

You can also adjust the heat curve and hence the indoor temperature by changing the "ROOM" setting. If the "ROOM" setting is used to adjust the heat curve it does not make the curve steeper or shallower, as happens when the "CURVE" value is changed. It instead shifts the entire heat curve through 3°C for each one degree change in the "ROOM" setting. The relationship between the supply line temperature and the outdoor temperature is unaffected, since the supply line temperature is increased or decreased by the same amount over the full length of the curve. In other words the entire heat curve is raised or lowered, instead of changing its slope.

This method of adjusting the indoor temperature should only be used to raise or lower the indoor temperature temporarily. If a long-term increase or decrease in the indoor temperature is required you should adjust the heat curve instead, as this is the most energy-efficient and cost-effective way of setting the heat curve (see "Adjusting the "CURVE" setting" on page 16).

The factory-set "ROOM" setting is 20°C.

Change the ROOM setting



To change the "ROOM" setting:

1 Press the up or down button once to select the "ROOM" setting so that you can change it.

2 Increase or decrease the "ROOM" setting using the up or down buttons to give the desired indoor temperature.

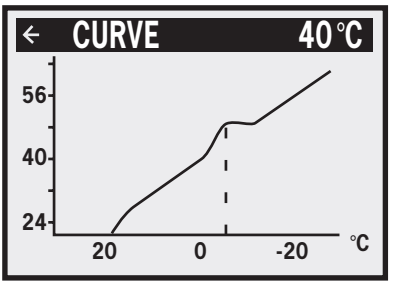
Wait 10 seconds or press the left button once to return to the character display's operating mode.

3.2.5 Adjusting a specific part of the heat curve

When the outdoor temperature is between –5°C and +5°C it may be necessary to adjust part of the heat curve if the indoor temperature does not remain constant at the set ROOM value. A facility is therefore provided to adjust the curve for three different outdoor temperatures: –5°C, 0°C and +5°C. This facility means that you can increase or decrease the supply line temperature at these three temperatures without affecting the rest of the heat curve. The supply line temperature is not just affected at the specified temperature, e.g. –5°C, but also progressively between 0°C and –10°C, so that the maximum adjustment is achieved at –5°C. The figure on the next page shows the adjusted CURVE –5. The adjustment can be seen as a bump in the graph.

You can choose to adjust the heat curve individually at three specific outdoor temperatures: –5°C, 0°C and +5°C. The supply line temperature can be changed by plus or minus 5 degrees.

Changing the value of CURVE –5



To change a specific part of the curve:

1 Press the right or left button once to open the main "INFORMATION" menu. The cursor will highlight the menu selection "OPERATION".

2 Press the down button to move the cursor over the menu selection "HEATCURVE".

3 Open the menu by pressing once on the right button. The cursor will highlight "CURVE".

4 Select "CURVE 5", "CURVE 0" or "CURVE –5" using the up or down buttons.

5 Open your selection by pressing the right button once.

6 Increase or decrease the setting using the up or down buttons.

Press the left button three times to return to the character display's operating mode.

3.2.6 Setting the "MIN" and "MAX" values

The supply line "MIN" and "MAX" temperatures are the lowest and highest values of the supply line temperature.

It is especially important to set the minimum and maximum supply line temperatures if you have underfloor heating.

If you have underfloor heating and parquet flooring in your house the supply line temperature should not exceed 45°C. Otherwise there is a risk that the parquet flooring could be damaged. If you have underfloor heating and stone flooring, the "MIN" setting should be adjusted to 22–25°C even in summer, when no heating is required. This is to ensure a comfortable floor temperature.

If your house has a cellar the "MIN" temperature should be also be set to suitable temperature in summer to prevent the cellar temperature from feeling cold. Remember that "HEATSTOP" also needs to be increased in order to provide heating in summer.

To change "MIN" or "MAX":

1 Press the right or left button once to open the main "INFORMATION" menu.
  The cursor will highlight the menu selection "OPERATION".

2 Press the down button to move the cursor over the menu selection "HEATCURVE".

3 Open the menu by pressing once on the right button. The cursor will highlight "CURVE".

4 Press the down button to move the cursor to "MIN".

5 Open your selection by pressing the right button once.
  The text "MIN" will be highlighted.

6 Increase or decrease the value using the up or down buttons.

7 Press the left button three times to return to the character display's operating mode.

Repeat the procedure to change the "MAX" value, but choose "MAX" instead of "MIN" in step 4.

3.2.7 Setting "HEATSTOP"

Heat stop is a function that switches off radiator heating completely when the outside temperature is equal to or higher than the set heat stop value.

When heat stop is activated, the circulation pump is switched off, except when producing warm water. The circulation pump does run for 1 minute/day to prevent seizing, however. The factory setting is 17°C.

To change "HEATSTOP":

1 Press the right or left button once to open the main "INFORMATION" menu. The cursor will highlight the menu selection "OPERATION".

2 Press the down button to move the cursor over the menu selection "HEATCURVE".

3 Open the menu by pressing once on the right button.

4 The cursor will highlight "CURVE".

5 Press the down button to move the cursor to "HEATSTOP".

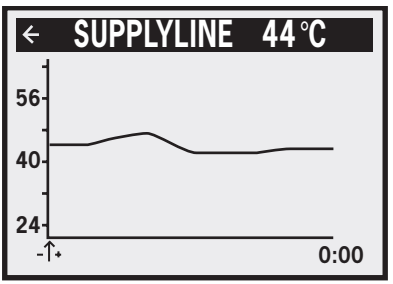
6 Open your selection by pressing the right button once. The text "HEATSTOP" will be highlighted.

7 Increase or decrease the value using the up or down buttons.

8 Press the left button three times to return to the character display's operating mode.

3.2.8 "TEMPERATURES" history

The installation can display a graph showing a history of the various temperatures so you can see how they have varied over the past hour. The example below shows how such a graph might look.



Histories are available for all temperatures except "ROOM", which is just the set value that is shown in the character display's operating mode. The integral value that can be displayed is the energy balance for the heating system.

To see the "TEMPERATURE" history:

1 Press the right or left button once to open the main "INFORMATION" menu.
The cursor will highlight the menu selection "OPERATION".

2 Press the down button to move the cursor over the menu selection "TEMPERATURE".

3 Open the menu by pressing once on the right button.

4 The cursor will highlight "OUTDOOR".

5 Press the the up or down button to move the cursor to the desired value.

6 Open your selection by pressing the right button once.
A graph will be drawn in the character display.

7 Move the cursor along the time axis using the up (plus) or down (minus) buttons.
A precise value for the selected time will be displayed at the top of the character display.

8 Press the left button three times to return to the character display's operating mode.

3.3 Maximum return line temperature

The maximum return line temperature, i.e. the temperature of the water that comes back from the heating system, is adapted to suit each individual installation. This will be set by your installer when the heat pump is installed, and can be adjusted by the installer at a later date.

3.4 Warm water production

The temperature of the warm water delivered to the warm water tank is controlled by the control pressostat and cannot be adjusted.

Reading the warm water temperature.

The current warm water temperature can be read from the character display by following these steps:

1 Press the right or left button once to open the main "INFORMATION" menu.
The cursor will highlight the menu selection "OPERATION".

2 Press the down button to move the cursor over the menu selection "TEMPERATURE".

3 Open the menu by pressing once on the right button.

4 Press the down button to move the cursor to "WARMWATER".

5 Open your selection by pressing the right button once.
A graph will be displayed showing the warm water temperature over the last hour.

6 Press the left button three times to return to the character display's operating mode.

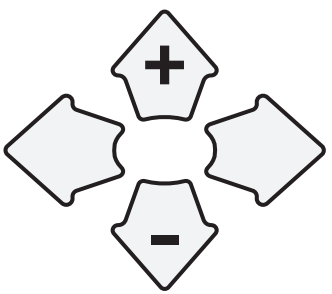
The value that is shown by the "WARMWATER" menu is the current warm water temperature, and the value in brackets is the temperature at which warm water production starts. When the temperature of the warm water drops below the start value the equipment will start producing warm water. The start value cannot be changed.

3.5 Regular checks

3.5.1 Check of Operating Mode

Under normal operating conditions, the alarm indicator will show a green fixed light which means that everything is in working order. In the event of an alarm, it will show a green blinking light and an alarm message will be displayed.







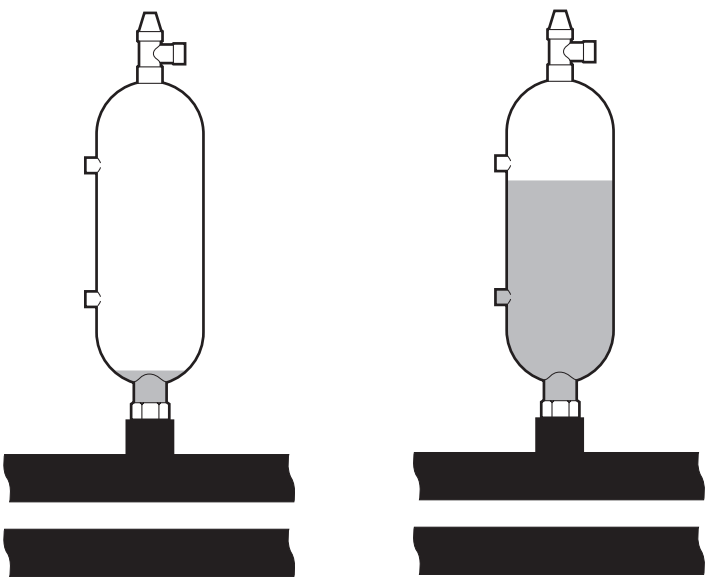
Blinking light indicating case of alarm

Check the alarm indicator regularly to make sure that the heat pump works properly. You would not always notice if there is something wrong, as the additional heater would start automatically (provided, of course, operating mode "AUTO" was selected). For more information, please refer to the chapter "TROUBLE SHOOTING" on p. 27.

3.5.2 Checking the coolant level

The coolant circuit must be filled with the correct amount of liquid, otherwise the installation may be damaged.

The coolant must be topped up when the level drops so far that it is no longer visible in the expansion tank.



Level too low

Level correct

In the first month after the installation is started up the coolant level may drop slightly, which is normal. The coolant level can also vary depending on the temperature of the heat source, but under no circumstances should the coolant level drop so far that it is no longer visible in the expansion tank.

Always contact your installer when the coolant requires topping up. (See "Reference information" on page 32).

3.5.3 Checking the water level in the heating system

The installation's system pressure should be checked once a month. The external pressure gauge should show a pressure of between 1–1.5 bar. If the value is less then 0.8 bar when the water in the heating system is cold then the system must be topped up with water (where closed expansion tank is fitted). See "Pipe installation (schematic layout)" on page 7 to find the location of the pressure gauge.

You can use normal tap water to top up the heating system. In some exceptional cases the quality of the tap water may be unsuitable for topping up the heating system (corrosive or hard water). Contact your installer if you are unsure. (See "Reference information" on page 32).



Do not add any water treatment chemicals to the heating system water!

3.5.4 Checking the safety valves

Both the heat pump's safety valves should be checked at least four times a year to prevent limescale deposits from clogging the mechanisms. See also "Pipe installation (schematic layout)" on page 7 to find out where the safety valves are located.

The safety valve on the warm water tank is intended to prevent excess pressure from building up in the closed warm water tank. It is mounted on the cold water inlet, with the outlet running downwards. If the safety valve on the warm water tank is not checked regularly there is a risk that the warm water tank could be damaged. It is normal for the safety valve to release small amounts of water while the warm water tank is being filled, especially when large quantities of warm water have been used.

Both safety valves can be tested by turning the cap a quarter turn clockwise and checking that the valves release a little water from the overflow pipe.

If either of the valves does not work they must be replaced. Contact your installer (see "Reference information" on page 32).

The opening pressures for the safety valves cannot be adjusted.

3.5.5 In case of leakage

If there is a leak from the warm water pipes between the heat pump installation and the taps, immediately close the shut-off valve for the cold water supply. Then contact your installer (see "Reference information" on page 32). If there is a leak from the refrigerant circuit, switch off the heat pump and contact your installer immediately. (See "Reference information" on page 32).

4 Troubleshooting

4.1 Alarms

If an error message appears, try restarting the heat pump by switching the circuit breaker on and off.

If restarting the heat pump does not work, try to solve the problem with the aid of Table 4. Contact your installer (see "Reference information" on page 32) if you cannot solve the problem yourself.

Table 4: Alarms

Message	Meaning	Cause	Remedy
LOWPRESS ERROR	Pressure is too low – The compressor is switched off and no warm water is produced.	Not enough antifreeze in the coolant system. Air in the coolant system. Blocked particle filter in coolant system.	Contact your installer.
HIGHPRESS ERROR	Pressure is too high – The compressor is switched off and no warm water is produced.	Radiator or underfloor heating thermostats are not turned up high enough. Air in heating system. Clogged particle filter in heating system.	Turn up radiator or underfloor heating thermostats. Top up and bleed heating system or contact your installer.
MOTOR P ERROR	Motor cut-out has tripped. The compressor is switched off and no warm water is produced.	Loss of power due to fuse failure or tripping of circuit breaker.	Contact your installer.
ALARM ADD.HEAT	Thermal cut-out tripped.	Electrical fault. Circuit breaker has tripped.	Contact your installer.
SENSOR OUTDOOR	Outdoor sensor fault. 0°C degrees is used to calculate set value for supply line temperature instead.	Electrical fault.	Contact your installer.
SENSOR SUPPLY	Supply line sensor fault. Everything is switched off except circulation pump for heating system.	Electrical fault.	Contact your installer.
SENSOR RETURN	Return line sensor fault.	Electrical fault.	Contact your installer.

Table 4: Alarms

Message	Meaning	Cause	Remedy
SENSOR WARMWATER	Warm water sensor fault. No warm water is produced.	Electrical fault.	Contact your installer.
ALARM ROOM SENSOR	Actual room temperature is not displayed. 20°C is used to calculate set supply line temperature instead.	Electrical fault.	Contact your installer.
ERR. PHASE SEQ.	Heat pump compressor is rotating in wrong direction, which means that heating is provided solely by additional heater.	Phase sequence has been reversed due to modification of the electrical installation in the house.	Contact your installer.
HIGH RETURN	The temperature of the water returning from the radiators is too high, which is causing a problem for the heat pump.	Radiator/underfloor heating thermostats not turned up high enough.	Make sure all thermostat valves are fully open.

4.1.0.1 Table 5 only applies to installations that use groundwater as the brine water.

Table 5: ***Only applies to Thermia Duo Optimum**

Message	Meaning	Cause	Remedy
BRINE UT	Brine temperature is lower than set value. Compressor is switched off and no warm water is produced.	The set minimum brine water temperature has been reached.	The system will reset itself when the temperature has risen to the set value.
BRINEFLOW LOW	The flow monitor was not active during the last start attempt. Compressor is switched off and no warm water is produced.	Brine water flow rate is low.	Contact your installer.
CIRC: PUMP*	The internal protection of the heating system's circulation pump has deployed.		Contact your installer.
BRINE: PUMP*	The internal protection of the coolant system's circulation pump has deployed.		Contact your installer.

5 Terms and abbreviations

Table 6: Terms and abbreviations

Evaporator	The evaporator extracts energy from the heat source, and the refrigerant that passes through the evaporator is transformed into its gaseous form. (See "How a heat pump works" on page 5 for more information).
"INTEGRAL"	The "INTEGRAL" is the energy balance of the heating system. Heat production is regulated by the calculated demand. This demand is calculated by comparing the actual supply line temperature with the calculated supply line temperature. The difference between the temperatures is multiplied by how long that difference exists. The result is called the integral. The integral is calculated automatically when the heating system is in operation. The value of the integral can be read from the character display under the "TEMPERATURE" menu.
Compressor	The compressor raises the temperature and pressure of the refrigerant. (See "How a heat pump works" on page 5 for more information).
Condenser	In the condenser the refrigerant gives off heat energy to the heating circuit. (See "How a heat pump works" on page 5 for more information).
"CURVE"	The "CURVE" value is set using the graphical character display. The set value determines the supply line temperature of the water that is delivered to the radiators when the outdoor temperature is 0°C.
Coolant	A water-based mixture that transfers energy from the heat source to the heat pump. (See "How a heat pump works" on page 5 for more information).
Coolant circuit	The liquid circuit that transfers heat from the heat source to the heat pump. (See "How a heat pump works" on page 5 for more information).
Refrigerant circuit	The circuit in the heat pump that transfers heat from the coolant circuit to the heating circuit by means of vaporization, compression and condensation. (See "How a heat pump works" on page 5 for more information).
Refrigerant	The liquid that transfers heat from the coolant circuit to the heating circuit. (See "How a heat pump works" on page 5 for more information).

Table 6: Terms and abbreviations

Radiator	Heating element.
Control computer	The control computer controls the entire heating installation. This is where all the settings are stored and where the installation history is recorded. The control computer settings are changed by means of a graphical character display.
"ROOM"	The "ROOM" setting shows the desired indoor temperature.
Sub-cooler	The sub-cooler preheats the return water from the radiator system and underfloor heating system. (See "How a heat pump works" on page 5 for more information).
Heating circuit	The heating circuit takes heat/energy from the refrigerant circuit and then transfers this to the warm water tank or radiator and underfloor heating systems. (See "How a heat pump works" on page 5 for more information).
Heat curve	The heat curve is the tool used by the control computer to calculate the supply line temperature required by the heating system. You adjust the indoor temperature by changing the slope of the heating system's "CURVE".

6 Initial settings for control computer

The first column in Table 7 shows the names of the adjustments that users can make themselves. The second column shows which values are set when the heat pump is delivered from the factory, and the third column shows the values set by the installer when your heat pump was installed.

As a user you should make sure that the installer fills in any special adjustments that were made for your particular heat pump, in order to make things easier when you come to make adjustments yourself.

Table 7: Initial settings

Setting	Factory setting	Setting specific to customer
ROOM	20°C	
OPERATION	AUTO	
CURVE	40°C	
MIN	10°C	
MAX	55°C	
CURVE 5	0°C	
CURVE 0	0°C	
CURVE -5	0°C	
HEATSTOP	17°C	
REDUCTION	3°C	

7 Reference information

Installation and commissioning have been carried out by:

Pipe installation	
Date:	
Company	
Name:	
Tel. no.:	

Electrical installation	
Date:	
Company	
Name:	
Tel. no.:	

Balancing of system	
Date:	
Company	
Name:	
Tel. no.:	

If these instructions are not followed during installation, operation, and maintenance, the obligations of Thermia Värme AB under the warranty regulations are no longer binding.

Thermia Värme AB reserves the right to make changes to parts and specification without prior notice.

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